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Lecture 27 3/30/26

Summary of last week:

- Driven-damped oscillator using Fourier analysis

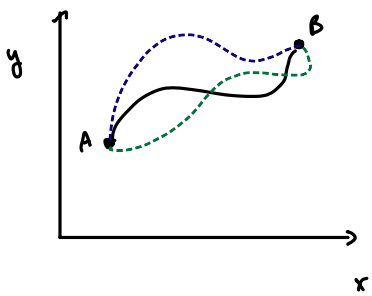
$$\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = f(t)$$

$$\hat{D} = \frac{d^2}{dt^2} + 2\beta \frac{d}{dt} + \omega_0^2 \quad \Rightarrow \quad \hat{D}\, x(t) = f(t)$$

- use Fourier series to write $f(t) = \sum_n f_n(t) = \sum_n a_n \cos(n\omega t)$
- solved for $Dx_n = a_n \cos(n\omega t)$ to find $x_n(t)$
- Derived $x(t) = \sum_n x_n(t)$

Friday Activity: Derived Euler-Lagrange Equations

Ch. 6: Calculus of variations

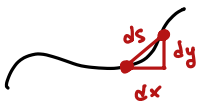


How can we find the "right" path?

Example: path that minimizes the distance between A & B

$$L = \int ds$$

$$ds^2 = d\vec{r} \cdot d\vec{r} \quad \Rightarrow \quad ds = \sqrt{dx^2 + dy^2}$$



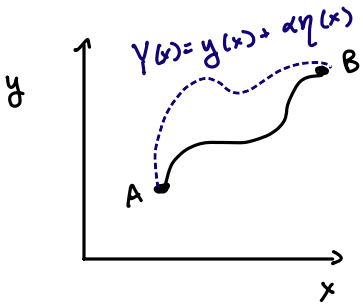
$$ds = \sqrt{(dx)^2 \left(1 + \frac{dy^2}{dx^2}\right)}$$

$$= dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$ds = \sqrt{1 + y'(x)^2}\, dx$$

$$\Rightarrow \quad L = \int \sqrt{1 + y'(x)^2}\, dx$$

TASK: Find $y(x)$ that <u>minimizes</u> L



$$Y(x) = y(x) + \alpha\, \eta(x)$$

α = parameter of how wrong Y is

When $\alpha = 0$, $Y = y$ = correct

$$S = \int_{x_A}^{x_B} f(Y, Y', x)\, dx$$

WTF y. s.t. $\left.\frac{\partial S}{\partial \alpha}\right|_{\alpha=0} = 0 \Rightarrow$ better be true if L is minimized

$$\frac{\partial S}{\partial \alpha} = \frac{\partial}{\partial \alpha} \int_{x_A}^{x_B} f(Y, Y', x)\, dx = \int_{x_A}^{x_B} \frac{\partial}{\partial \alpha} \underbrace{f(Y, Y', x)}_{\text{use chain rule}}\, dx$$

$$\frac{\partial f}{\partial \alpha}(Y, Y', x) = \frac{\partial f}{\partial Y}\frac{\partial Y}{\partial \alpha} + \frac{\partial f}{\partial Y'}\frac{\partial Y'}{\partial \alpha} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial \alpha}$$

} Chain rule for multivariable fns

$$\frac{\partial Y}{\partial \alpha} = \frac{\partial}{\partial \alpha}(y + \alpha\eta) = \eta$$

$$\frac{\partial Y'}{\partial \alpha} = \frac{\partial}{\partial \alpha}(y' + \alpha\eta') = \eta'$$

$$\frac{\partial f}{\partial \alpha}(Y, Y', x) = \frac{\partial f}{\partial Y}\eta + \frac{\partial f}{\partial Y'}\eta' + 0$$

$$\left.\frac{\partial f}{\partial \alpha}\right|_{\alpha=0} = \frac{\partial f}{\partial y}\eta + \frac{\partial f}{\partial y'}\eta' \qquad \left[\, Y|_{\alpha=0} = y \,\right]$$

$$\underbrace{\frac{\partial S}{\partial \alpha} = \int_{x_A}^{x_B} \frac{\partial f}{\partial \alpha}\, dx}_{\text{at } \alpha=0} = \int_{x_A}^{x_B} \left[\eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'}\right] dx = 0$$

$$= \int_{x_A}^{x_B} \eta \frac{\partial f}{\partial y}\, dx + \int_{x_A}^{x_B} \left[\frac{d}{dx}\eta\right] \frac{\partial f}{\partial y'}\, dx = 0$$

$dv = \eta' dx$ $\quad u = \frac{\partial f}{\partial y'}$

$v = \eta$ $\quad du = \frac{d}{dx}\frac{\partial f}{\partial y'}$

$$= \int_{x_A}^{x_B} \eta \frac{\partial f}{\partial y}\, dx + \left.\eta \frac{\partial f}{\partial y}\right|_{x_A}^{x_B} - \int_{x_A}^{x_B} \eta \frac{d}{dx}\frac{\partial f}{\partial y'}\, dx$$

$$= \int_{x_A}^{x_B} \eta \left[\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'}\right] dx = 0 \Rightarrow \boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0}$$

Euler-Lagrange Equations!

Apply this to our example:

$$L = \int ds = \int \sqrt{1 + y'(x)}\, dx$$

$$f = \sqrt{1 + y'(x)^2}$$

$$\frac{\partial f}{\partial y} = 0 \qquad \frac{\partial f}{\partial y'} = \frac{1}{2}\left[1 + (y')^2\right]^{-1/2} 2y' \Rightarrow \frac{\partial f}{\partial y} = \frac{d}{dx}\frac{\partial f}{\partial y'}$$

$$= \frac{y'(x)}{\sqrt{1 + y'(x)^2}}$$

$$0 = \frac{d}{dx}\frac{y'(x)}{\sqrt{1 + y'(x)^2}}$$

$$\Rightarrow C = \frac{y'(x)}{\sqrt{1 + y'(x)^2}}$$

$$c = \frac{y'(x)}{\sqrt{1+y'(x)^2}}$$

$$c^2 = \frac{y'(x)^2}{1+y'(x)^2}$$

$$c^2 + c^2 y'(x)^2 = y'(x)^2$$

$$c^2 = (1-c^2)\, y'(x)^2$$

$$\frac{c}{\sqrt{1-c^2}} = y'(x)$$

let $m \equiv \frac{c}{\sqrt{1-c^2}}$

a bit tedious but demonstrates the point

$$m = y'(x)$$

$$\int_0^x m\, dx' = \int_0^x y'(x')\, dx'$$

$$mx = y - y(0)$$

$$mx + y(0) = y$$

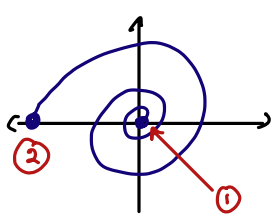
$$\boxed{y = mx + b}$$ where $b \equiv y(0)$

Note: Euler-Lagrange equations only guarantee:

$$\left.\frac{\partial S}{\partial \alpha}\right|_{\alpha=0} = 0 \Rightarrow$$ stationary point "critical point" doesn't specify max or min

It's difficult to prove you've minimized S in general, but for particular examples it can be easy to verify.

More than two variables



write path as a parametric function $x = x(u)$ $y = y(u)$

Note: u is often time in physics problems

The shortest distance exampe again:

$$L = \int ds \quad , \quad ds = \sqrt{dx^2 + dy^2} = \sqrt{\frac{dx^2}{du^2} + \frac{dy^2}{du^2}}\, du = \sqrt{x'(u)^2 + y'(u)^2}\, du$$

$$L = \int_{u_1}^{u_2} \sqrt{x'(u)^2 + y'(u)^2}\, du$$

$$S = \int_{u_1}^{u_2} f(x, y, x', y', u)\, du$$

$$X = x(u) + \alpha \xi(u)$$

$$Y = y(u) + \beta \eta(u)$$

Stationary point $\frac{\partial S}{\partial \alpha} = \frac{\partial S}{\partial \beta} = 0$

By the same argument as before:

$$\frac{\partial f}{\partial x} = \frac{d}{du} \frac{\partial f}{\partial x'} \qquad \frac{\partial f}{\partial y} = \frac{d}{du} \frac{\partial f}{\partial y'}$$

Derivation with x & y:

$$\frac{\partial S}{\partial \alpha} = 0 = \int_{u_1}^{u_2} \left(\xi \frac{\partial f}{\partial x} + \xi' \frac{\partial f}{\partial x'} \right) du = \int_{u_1}^{u_2} \left(\xi \frac{\partial f}{\partial x} - \xi \frac{d}{du} \frac{\partial f}{\partial x'} \right) du = \int_{u_1}^{u_2} \xi \left(\frac{\partial f}{\partial x} - \frac{d}{du} \frac{\partial f}{\partial x'} \right) du$$

$$\Rightarrow \frac{\partial f}{\partial x} - \frac{d}{du} \frac{\partial f}{\partial x'} = 0$$

$$\frac{\partial S}{\partial \beta} = 0 = \int_{u_1}^{u_2} \left(\eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'} \right) du = \int_{u_1}^{u_2} \left(\eta \frac{\partial f}{\partial y} - \eta \frac{d}{du} \frac{\partial f}{\partial y'} \right) du = \int_{u_1}^{u_2} \eta \left(\frac{\partial f}{\partial y} - \frac{d}{du} \frac{\partial f}{\partial y'} \right) du$$

$$\Rightarrow \frac{\partial f}{\partial y} - \frac{d}{du} \frac{\partial f}{\partial y'} = 0$$

Example to minimize distance b.t. ① & ②:

$$L = \int_{u_1}^{u_2} \sqrt{x'(u)^2 + y'(u)^2}\, du$$

$$\frac{\partial f}{\partial x} = 0 \qquad \frac{\partial f}{\partial x'} = \frac{1}{2} \left[(x')^2 + (y')^2 \right]^{-1/2} 2x' = \frac{x'}{\sqrt{(x')^2 + (y')^2}}$$

$$\frac{\partial f}{\partial y} = 0 \qquad \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{(x')^2 + (y')^2}}$$

$$\frac{\partial f}{\partial x} = \frac{d}{du}\frac{\partial f}{\partial x'} \qquad \frac{\partial f}{\partial y} = \frac{d}{du}\frac{\partial f}{\partial y'}$$

$$0 = \frac{d}{du}\frac{x}{\sqrt{(x')^2+(y')^2}} \qquad 0 = \frac{d}{du}\frac{y'}{\sqrt{(x')^2+(y')^2}}$$

$$C_1 = \frac{x}{\sqrt{(x')^2+(y')^2}} \qquad C_2 = \frac{y'}{\sqrt{(x')^2+(y')^2}}$$

$$\frac{C_1}{C_2} = \frac{x'}{y'} \doteq m \qquad y' = mx'$$

$$\frac{dy}{\cancel{du}} = m\frac{dx}{\cancel{du}}$$

$$\boxed{y = mx + b}$$

In general, we will have many coordinates:

$$\mathcal{L}(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$$

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)\, dt \Rightarrow n \text{ different Euler-Lagrange equations}$$

$$\frac{\partial \mathcal{L}}{\partial q_n} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_n}$$ all of the same form.

Chapter 7: Lagrange's Equations

Consider a particle moving in 3D, subject to $\vec{F}(\vec{r})$, where $\vec{F}$ is conservative.

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\,\dot{\vec{r}}\cdot\dot{\vec{r}} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

$$U = U(\vec{r}) = U(x, y, z)$$

We can define a <u>Lagrangian</u> function:

$$\mathcal{L} = T - U$$

$\mathcal{L}$ determines the motion of the system in a very general way. You can think of $\mathcal{L}$ as a quantity that balances T & U.